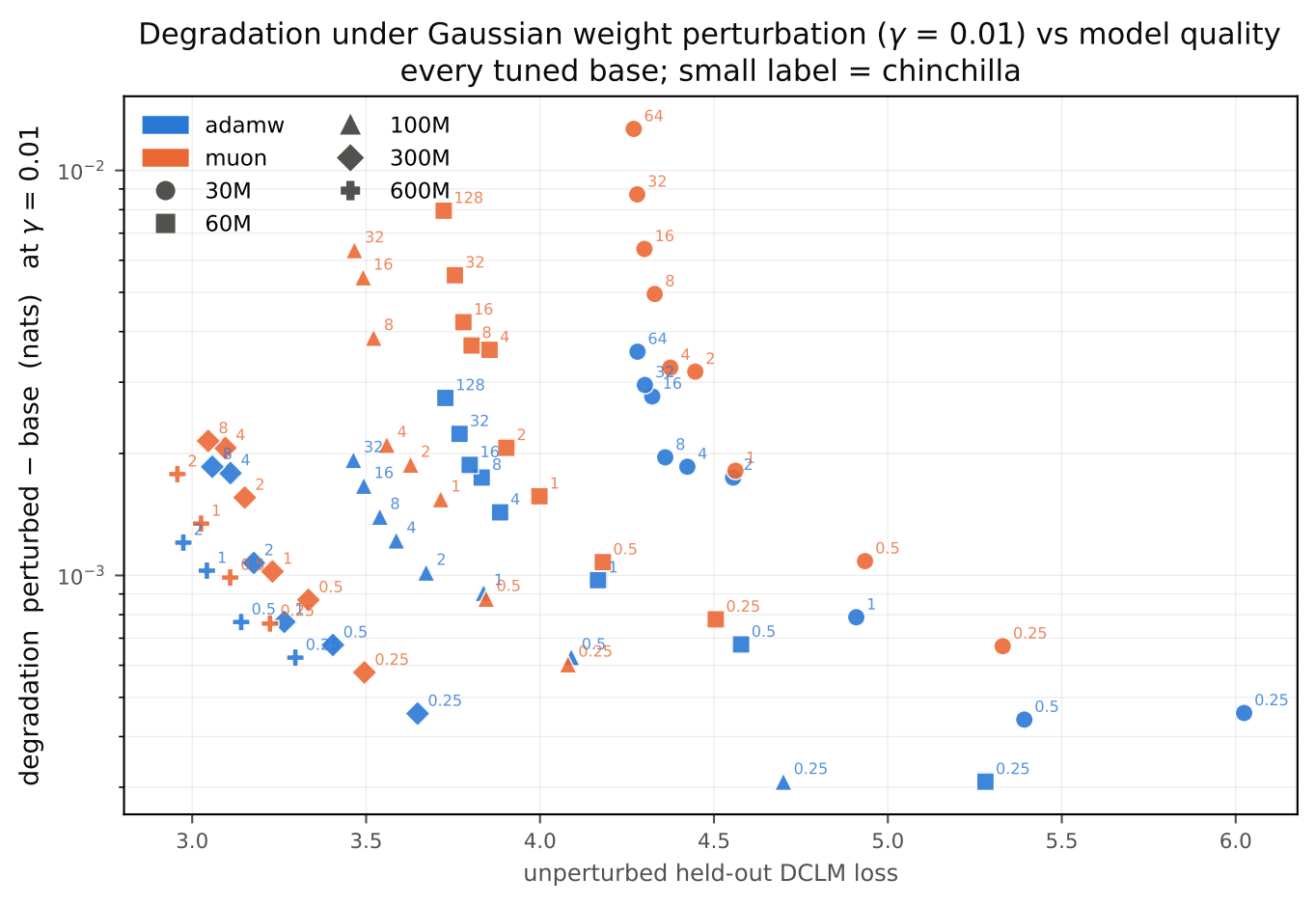
Degradation under Gaussian weight perturbation (γ = 0.01) vs model quality
every tuned base; small label = chinchilla
adamw
muon
30M
60M
100M
300M
600M
degradation perturbed − base (nats) at γ = 0.01
10^{-2}
10^{-3}
3.0
3.5
4.0
4.5
5.0
5.5
6.0
unperturbed held-out DCLM loss